



# PX-58: Pre-Alloyed Mo-Fe + C + AncorCut + Lubricant

## Safety Data Sheet

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

Date of issue: 08/02/2018

Revision date: 03/28/2019

Version: 1.1

### 1.1. Identification

Product form : Mixture

Product name : PX-46: Pre-Alloyed Mo-Fe + C + AncorCut + Lubricant

### 1.2. Recommended use and restrictions on use

Use of the substance/mixture : Powder Metallurgy Use

### 1.3. Supplier

Hoeganaes Corporation  
1315 Airport Road  
Gallatin, TN 37066 - USA  
T: (615) 451-2000

Hoeganaes Corporation  
4330 Paradise Road  
Watson, PA 17777 - USA  
T: (570) 538-3587

ARC Metals, Hoeganaes Corporation  
PO Box 372  
224 River Road  
Ridgway, PA 15853 - USA T:  
(814) 776-2116  
F: (814) 776-4000

### 1.4. Emergency telephone number

Emergency number : INFOTRAC Emergency Response Center  
200 North Palmetto Street; Leesburg, Florida 34748  
Tel.# 800-535-5053 / 352-323-3500  
Fax.# 352-326-3902

## SECTION 2: Hazard(s) identification

### 2.1. Classification of the substance or mixture

#### GHS-US classification

Combustible Dust

### 2.2. GHS Label elements, including precautionary statements

#### GHS-US labeling

Signal word (GHS-US) : Warning

Hazard statements (GHS-US) : May form combustible dust concentrations in air

### 2.3. Other hazards which do not result in classification

No additional information available

### 2.4. Unknown acute toxicity (GHS US)

Not applicable

## SECTION 3: Composition/Information on ingredients

### 3.1. Substances

Not applicable

### 3.2. Mixtures

| Name        | Product identifier  | %    |
|-------------|---------------------|------|
| Iron        | (CAS-No.) 7439-89-6 | ≥ 92 |
| Graphite ** | (CAS-No.) 7782-42-5 | < 2  |

\*Chemical name, CAS number and/or exact concentration have been withheld as a trade secret

### 3.3. Hazardous components

\*\* Per XRC(M) analysis, which combines the analytical capabilities of X-Ray Diffraction/Fluorescence, Computer Controlled Scanning Electron Microscopy/Energy Dispersive Spectroscopy, Raman Spectroscopy and Inductively Coupled Plasma to conduct particle-by-particle intrainstrumental physicochemical/mineralogical analysis – any naturally occurring Respirable Crystalline Silica (RCS) that may exist in this product is inextricably bound, environmentally unavailable and at de minimis concentration. Thus, in its current and anticipated future physical state, the product is incapable of causing toxicologically relevant RCS exposure under either normal conditions of use or in an extreme upset scenario.

: Pre-Alloyed Mo-

+ Lubricant

## SECTION 4: First-aid measures

according to the Hazard Communication Standard (CFR29 1910.1200) HazCom 2012.

#### 4.1. Description of first aid measures

|                                       |                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| First-aid measures after inhalation   | : If breathing is difficult, remove person to fresh air and keep at rest in a position comfortable for breathing. Get medical advice/attention if you feel unwell.                           |
| First-aid measures after skin contact | : If skin irritation occurs: Wash skin with plenty of water. Obtain medical attention if irritation persists.                                                                                |
| First-aid measures after eye contact  | : IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. |
| First-aid measures after ingestion    | : Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person. Get medical advice/attention if you feel unwell.                                     |

#### 4.2. Most important symptoms and effects (acute and delayed)

|                                     |                                                                                                                                               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Symptoms/effects after inhalation   | : May cause irritation to the respiratory tract.                                                                                              |
| Symptoms/effects after skin contact | : May cause skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin.                                        |
| Symptoms/effects after eye contact  | : May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling. |
| Symptoms/effects after ingestion    | : May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.                                          |

#### 4.3. Immediate medical attention and special treatment, if necessary

Symptoms may be delayed. In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

### SECTION 5: Fire-fighting measures

#### 5.1. Suitable (and unsuitable) extinguishing media

|                                |                                                                 |
|--------------------------------|-----------------------------------------------------------------|
| Suitable extinguishing media   | : Use extinguishing media appropriate for surrounding fire.     |
| Unsuitable extinguishing media | : Do not use a water jet since it may cause the fire to spread. |

#### 5.2. Specific hazards arising from the chemical

|                  |                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------|
| Fire hazard      | : Combustible dust. Products of combustion may include, and are not limited to: oxides of carbon.                   |
| Explosion hazard | : Airborne dust in sufficient concentrations when confined and exposed to a sufficient ignition source can explode. |
| Reactivity       | : No dangerous reactions known under normal conditions of use.                                                      |

#### 5.3. Special protective equipment and precautions for fire-fighters

Protection during firefighting : Keep upwind of fire. Wear full fire fighting turn-out gear (full Bunker gear) and respiratory protection (SCBA).

### SECTION 6: Accidental release measures

#### 6.1. Personal precautions, protective equipment and emergency procedures

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General measures | : Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel. Dust deposits should not be allowed to accumulate on surfaces, as these may form an explosive mixture if they are released into the atmosphere in sufficient concentration. Avoid dispersal of dust in the air (i.e., clearing dust surfaces with compressed air). Consider the use of non-sparking tools. |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### 6.1.1. For non-emergency personnel

No additional information available

##### 6.1.2. For emergency responders

No additional information available

#### 6.2. Environmental precautions

Prevent entry to sewers and public waters.

#### 6.3. Methods and material for containment and cleaning up

For containment : Contain spill, then place in a suitable container. Minimize dust generation. Do not flush to sewer or allow to enter waterways. Use appropriate Personal Protective Equipment (PPE). Methods for cleaning up : Vacuum or sweep material and place in a disposal container. Provide ventilation.

#### 6.4. Reference to other sections

For further information refer to section 8: "Exposure controls/personal protection"

**: Pre-Alloyed Mo-**

**+ Lubricant**

**SECTION 7: Handling and storage****7.1. Precautions for safe handling**

Precautions for safe handling : Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Avoid contact with skin and eyes. Avoid generating and breathing dust. Do not swallow. Handle and open container with care. When using do not eat, drink or smoke. Good housekeeping is important to prevent accumulation of dust. The use of compressed air for cleaning clothing, equipment, etc, is not recommended. Use only in well ventilated areas. Handling this product may result in electrostatic accumulation. Use proper grounding procedures.

Hygiene measures : Wash contaminated clothing before reuse. Wash hands, forearms and face thoroughly after handling.

**7.2. Conditions for safe storage, including any incompatibilities**

Storage conditions : Keep away from sources of ignition. Keep out of the reach of children. Store in dust-tight, dry, labelled containers. Avoid any dust buildup by frequent cleaning and suitable construction of the storage area.

**SECTION 8: Exposure controls/personal protection****8.1. Control parameters****Iron (7439-89-6)**

Not applicable

**Graphite (7782-42-5)**

|       |                                      |                                                                                                    |
|-------|--------------------------------------|----------------------------------------------------------------------------------------------------|
| ACGIH | ACGIH TWA (mg/m <sup>3</sup> )       | 2 mg/m <sup>3</sup> (all forms except graphite fibers-respirable particulate matter)               |
| OSHA  | OSHA PEL (TWA) (mg/m <sup>3</sup> )  | 15 mg/m <sup>3</sup> (synthetic-total dust)<br>5 mg/m <sup>3</sup> (synthetic-respirable fraction) |
| IDLH  | US IDLH (mg/m <sup>3</sup> )         | 1250 mg/m <sup>3</sup>                                                                             |
| NIOSH | NIOSH REL (TWA) (mg/m <sup>3</sup> ) | 2.5 mg/m <sup>3</sup> (natural-respirable dust)                                                    |

**8.2. Appropriate engineering controls**

Appropriate engineering controls : Ensure good ventilation of the work station. It is recommended that all dust control equipment such as local exhaust ventilation and material transport systems involved in handling of this product contain explosion relief vents or an explosion suppression system or an oxygen-deficient environment. Ensure that dust-handling systems (such as exhaust ducts, dust collectors, vessels, and processing equipment) are designed in a manner to prevent the escape of dust into the work area (i.e, there is not leakage from the equipment) and are properly grounded. Use only appropriately classified electrical equipment and powered industrial trucks.

Environmental exposure controls : Avoid release to the environment. Maintain levels below Community environmental protection thresholds.

**8.3. Individual protection measures/Personal protective equipment****Hand protection:** Wear

suitable gloves **Eye**

**protection:**

Safety glasses or goggles are recommended when using product.

**Skin and body protection:**

Wear suitable protective clothing

**Respiratory protection:**

In case of insufficient ventilation, wear suitable respiratory equipment. Respirator selection must be based on known or anticipated exposure levels,

**: Pre-Alloyed Mo-**

**+ Lubricant**

## Safety Data Sheet

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the hazards of the product and the safe working limits of the selected respirator.

**Other information:**

Do not eat, drink or smoke when using this product. Always wash hands after handling the product. Handle in accordance with good industrial hygiene and safety procedures.

**SECTION 9: Physical and chemical properties****9.1. Information on basic physical and chemical properties**

|                                             |                          |
|---------------------------------------------|--------------------------|
| Physical state                              | : Solid                  |
| Appearance                                  | : Powder                 |
| Color                                       | : Gray                   |
| Odor                                        | : None                   |
| Odor threshold pH                           | : No data available      |
| Melting point                               | : No data available      |
| Freezing point Boiling point                | : 2798 °F (1536 °C)      |
| Flash point                                 | : No data available      |
| Relative evaporation rate (butyl acetate=1) | : No data available      |
| Flammability (solid, gas)                   | : No data available      |
| Vapor pressure                              | : No data available      |
| Relative vapor density at 20 °C             | : No data available      |
| Density                                     | : No data available      |
| Solubility                                  | : 7.36 g/cm <sup>3</sup> |
| Partition coefficient n-octanol/water       | : Insoluble in water     |
| Auto-ignition temperature                   | : No data available      |
| Decomposition temperature                   | : No data available      |
| Viscosity, kinematic                        | : No data available      |
| Viscosity, dynamic                          | : No data available      |
| Explosion limits                            | : No data available      |
| Explosive properties                        | : No data available      |

**9.2. Other information**

|             |      |
|-------------|------|
| VOC content | : 0% |
|-------------|------|

**SECTION 10: Stability and reactivity****10.1. Reactivity**

No dangerous reactions known under normal conditions of use.

**10.2. Chemical stability**

Stable under normal conditions. May form combustible dust concentrations in air.

**10.3. Possibility of hazardous reactions**

No dangerous reactions known under normal conditions of use.

**10.4. Conditions to avoid**

Keep away from heat, sparks and flames. Avoid dust formation. Incompatible materials.

**10.5. Incompatible materials**

Strong oxidizing agents. Strong acids.

**10.6. Hazardous decomposition products**

May include, and are not limited to: oxides of carbon.

**SECTION 11: Toxicological information****11.1. Information on toxicological effects**

|                |                  |
|----------------|------------------|
| Acute toxicity | : Not classified |
|----------------|------------------|

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| Iron (7439-89-6)                                   |                                                                                                                                               |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| LD50 oral rat                                      | 30 g/kg                                                                                                                                       |
| ATE US (oral)                                      | 30000 mg/kg bodyweight                                                                                                                        |
| Skin corrosion/irritation                          | : Not classified                                                                                                                              |
| Serious eye damage/irritation                      | : Not classified                                                                                                                              |
| Respiratory or skin sensitization                  | : Not classified                                                                                                                              |
| Germ cell mutagenicity                             | : Not classified                                                                                                                              |
| Carcinogenicity                                    | : Not classified                                                                                                                              |
| Reproductive toxicity                              | : Not classified                                                                                                                              |
| Specific target organ toxicity – single exposure   | : Not classified                                                                                                                              |
| Specific target organ toxicity – repeated exposure | : Not classified                                                                                                                              |
| Aspiration hazard                                  | : Not classified                                                                                                                              |
| Symptoms/effects after inhalation                  | : May cause irritation to the respiratory tract.                                                                                              |
| Symptoms/effects after skin contact                | : May cause skin irritation. Symptoms may include redness, drying, defatting and cracking of the skin.                                        |
| Symptoms/effects after eye contact                 | : May cause eye irritation. Symptoms may include discomfort or pain, excess blinking and tear production, with possible redness and swelling. |
| Symptoms/effects after ingestion                   | : May be harmful if swallowed. May cause gastrointestinal irritation, nausea, vomiting and diarrhea.                                          |
| Other information                                  | : Likely routes of exposure: ingestion, inhalation, skin and eye.                                                                             |

### SECTION 12: Ecological information

No additional information available

### SECTION 13: Disposal considerations

#### 13.1. Disposal methods

Product/Packaging disposal recommendations : Dispose of contents/container in accordance with local, regional, national and/or international regulations. The generation of waste should be avoided or minimized wherever possible.

### SECTION 14: Transport information

#### Department of Transportation (DOT)

In accordance with DOT

Not regulated

### SECTION 15: Regulatory information

#### 15.1. US Federal regulations

All components of this product are listed, or excluded from listing, on the United States Environmental Protection Agency Toxic Substances Control Act (TSCA) inventory.

#### 15.2. International regulations

No additional information available

#### 15.3. US State regulations

California Proposition 65 - Based on results of testing described in Section 3.3, and in accordance with UN-GHS and Prop-65 regulations, this material may be exempt from classification and labeling due to the absence of toxicologically relevant Respirable Si-Rich particulate.

### SECTION 16: Other information

Date of issue : 08/02/2018 Revision date : 03/28/2019

Other information : Refer to NFPA 654, Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids, for safe handling.

# PX-58

## Safety Data Sheet

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**PX-58**

Safety Data Sheet

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EN (English US)